

Comp 4531 Project Proposal

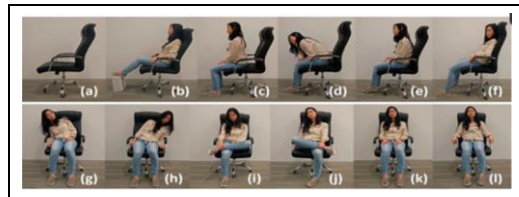
Real time Action Recognition with ML

Team members:

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Project Description

Sitting a long time to work or study, we may not be aware of our sitting habits that could seriously affect our spine. Having Elder alone at home, he/ she may unfortunately fall and no one aware of the serious situation. All these lead to the potential usage of action recognition in home or office to either keep track of our behaviors so as to improve our health, safety and work promptly. **Datasets** will be combining available databases online such as Kaggle, and include from sitting, sleeping, standing, etc.



Functions:

- 1) Real-time keep track of our study/ work habits/ sleeping habits/ elderly
- 2) Convert to human skeleton data (privacy)
- 3) Display any alerts on mobile phone (individual or caregivers)

Software requirements:

- Colab (training model)
- ML Models: real time lightweight pretrained model (e.g. YOLOv8, MobileNet)
- Deploy in Azure or similar cloud (optional)

Hardware requirements:

- Raspberry Pi: data processing and collection
- Camera module: capture video with suitable frame rate (Pi Camera)
- LoRaWAN or BLE: to communicate and store data in mobile phone
- Any other related sensors as personal assistance (temperature, humidity)